Jayesh Ingale

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Education

North Carolina State University — Raleigh, NC
Master of Science, Electrical and Computer Engineering | ***GPA**: 3.76/4.0* — *August 2024 – May 2026*

Coursework: Gen AI: Applications, Computer Vision, Pattern Recognition & Machine Learning, Cloud Computing.

College of Engineering Pune — India
Bachelor of Technology in Electrical Engineering | ***GPA**: 8.08/10* — *August 2018 – May 2022*

Technical Skills

Languages & Backend: Python, C++, SQL, TypeScript, FastAPI, Streamlit, PostgreSQL, SQLite, Redis, Airflow, dbt, ETL/ELT, Jira
Generative AI & LLM: Transformers, Hugging Face, BERT, LoRA/QLoRA, PEFT, DPO, Prompt Engineering, RAG, LangChain, LangGraph, MCP, Guardrails, FAISS, ChromaDB, Groq, vLLM, Ollama, Claude, Ragas, promptfoo, LangSmith, pytest
ML & MLOps: PyTorch, Scikit-learn, XGBoost, NumPy, Pandas, ONNX, Triton, MLflow, Docker, Kubernetes, GitHub Actions, AWS

Experience

Early Stage Startup – Precision Sustainable Ag. — Raleigh, NC
Applied ML Engineer — *June 2026 – Present*

- Integrating **stereovision** and **monocular** Depth Pro maps for **depth** completion into the current pipeline, adding the feature to MTL model.

Applied ML Intern — *May 2025 – May 2026*

- **Fine-tuned** DeepLabV3+ with ResNet-101 encoders in **PyTorch** to develop a **multitask learning** (**MTL**) model for semantic segmentation and regression, optimized with a **weighted multi-task loss** to balance task objectives.
- Built an end-to-end machine learning pipeline leveraging multiple NVIDIA A100 **GPU**s, reducing training time by **43%**.
- Deployed models with **MLflow**, enabled experiment tracking, and automated CI workflows using **GitHub Actions**.
- Implemented **ONNX** model serving with **NVIDIA Triton** Inference Server, leveraging dynamic batching and multi-instance execution to optimize GPU utilization and improve inference **throughput** by **20x**.
- Applied **Causal Reasoning** to measure how surrounding vegetation (**ATE**) drives clover prediction error, tracing the error to occlusion.
- Applied **Explainable AI** (**Grad-CAM**) to rule out spurious correlations, validating model behavior for domain experts and stakeholders.

Deloitte – AI & Data — India
Data Scientist — *January 2023 – July 2024*

- Prototyped an internal documentation assistant for an insurance client, leveraging **Retrieval Augmented Generation** and prompt engineering on **AWS** (**Bedrock**, Amazon **Titan** embeddings, **AWS Knowledge Base**, **S3**), reducing query response time by 30%.
- Designed **ELT data pipelines (Python, SQL, database systems)** with Data Engineering team to consolidate data across **10**+ services into single source of truth (**SSOT**), improving data reliability and handling 1M+ records monthly in production.

Projects

Self Evolving Topic Ontology for Earnings Calls | Link
Stack: LangGraph, LangSmith, MCP, Groq, ChromaDB, SQLite, FASTAPI, TypeScript

- Reproduced Bloomberg's "**Agentic Retrieval of Topic and Insights from Earnings Calls**", ingesting transcripts into self evolving topic ontology persisted in **SQLite/ChromaDB**.
- Orchestrated Supervisor, Retriever, Ontologist, RAG and Judge agents in LangGraph over **rate limited Groq inference**, with citation enforcing verification loop.
- Built three evaluation suites, consisting of ontology coherence(0.48 vs 0.18 baseline), **retrieval hit@10** of 0.80, and **LLM-as-judge** faithfulness, along with LangSmith tracing.
- Shipped a streaming FastAPI backend, React/TS frontend, and **Model Context Protocol (MCP)** server exposing it to external clients.

Self hosted AI Banking Copilot with Layered LLM Guardrails | Link
Stack: Llama 3.2, LangGraph, FASTAPI, PostgreSQL/pgvector, Presidio

- Built an **on-premise** copilot that lets bank managers query customer portfolios, using **text-to-SQL** and vector RAG over PostgreSQL.
- Engineered **guardrail** pipeline using Presidio **PII** redaction, **prompt-injection** detection, AST-level SQL validation, **human-in-the-loop** approval, and SHA-256 hash-chained audit logging.
- Tested with a 30-case **red-team** suite and unseen attack sets, achieving 85% precision with guardrails adding only 5% latency overhead.

Hybrid Retrieval and NLI verification for Legal Contracts | Link
Stack: FastAPI, Streamlit, LangGraph, FAISS, MCP, Langfuse, promptfoo/pytest CI

- Built citation grounded QA over 510 **CUAD** contracts, linking every Mistral generated claim with source text to trace hallucinations.
- Engineered **hybrid retrieval** using **BM25**, **FAISS** embeddings, with **reciprocal rank fusion** and **cross encoder reranking**.
- Added **NLI model** that verifies each claim against its citation and rewrites unsupported ones, 0.875 accuracy on unseen **LegalBench** corpus.
- Designed evaluation harness scoring **faithfulness** with dual verification, NLI(0.42) and LLM-judge(0.78) to quantify metric reliability.

[LLM Fine-tuning] LLM Alignment using Direct Preference Optimization | Link

- Aligned Qwen2.5-7B-Instruct using **DPO** with **PEFT** via **LoRA** to learn responses on ethical questions around AI-generated art.
- Leveraged Unsloth to enable 4-bit **quantization** and optimized inference, and leveraged an **LLM-as-a-judge** framework to compare pre- vs. post-alignment outputs.

[Computer Vision] Self-supervised Contrastive Learning | Link

- Trained SSL models in **PyTorch**, with **SimCLR** and Barlow Twins methods on the **STL10** dataset, achieving **top-1** validation accuracy of **62.17%** and **59.46%**, respectively.
- Conducted a **logistic regression** linear evaluation on embeddings across 10–500 images per class, observing a relative gain of about 50%, and **benchmarking** against a supervised **baseline** achieving 58.35%.